



Geographical distribution and extinction risk: lessons from Triassic–Jurassic marine benthic organisms

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ABSTRACT

Aim To evaluate the influence of geographical distribution on the extinction risk of benthic marine invertebrates using data from the fossil record, both during times of background extinction and across a mass-extinction episode. Total geographical range is contrasted with proxies of global abundance to assess the relationships between the two essential components of geographical distribution and extinction risk.

Location A global occurrence data base of fossil benthic macro-organisms from the Triassic and Jurassic periods was used for this study.

Methods Geographical distributions and biodiversity dynamics were assessed for each genus (all taxa) or species (bivalves) based on a sample-standardized data set and palaeogeographical reconstructions. Geographical ranges were measured by the maximum great circle distance of a taxon within a stratigraphic interval. Global abundance was assessed by the number of localities at which a taxon was recorded. Widespread and rare taxa were separated using median and percentile values of the frequency distributions of occurrences.

Results The frequency distribution of geographical ranges is very similar to that for modern taxa. Although no significant correlation could be established between local abundance and geographical range, proxies of global abundance are strongly correlated with geographical range. Taxon longevities are correlated with both mean geographical range and mean global abundance, but range size appears to be more critical than abundance in determining extinction risk. These results are valid when geographical distribution is treated as a trait of taxa and when assessed for individual geological stages.

Main conclusions Geographical distribution is a key predictor of extinction risk of Triassic and Jurassic benthic marine invertebrates. An important exception is in the end-Triassic mass extinction, which equally affected geographically restricted and widespread genera, as well as common and rare genera. This suggests that global diversity crises may curtail the role of geographical distribution in determining extinction risk.

Keywords

Abundance, background extinction, geographical range, Jurassic, marine invertebrates, mass extinction, origination, Triassic.

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INTRODUCTION

Geographical distribution is often regarded as a key determinant of extinction risk, in both recent and fossil taxa, even if a number of additional traits are also linked to extinction risk (McKinney, 1997). Although a few, mostly palaeontological

studies see no predictable relationship between geographical distribution and extinction risk (Stanley, 1986; Stanley *et al.*, 1988; Norris, 1992), most studies on modern taxa support the view that geographically widespread and common species are less endangered than geographically restricted and rare species (Johnson, 1998; Purvis *et al.*, 2000; Fagan *et al.*, 2002; Jones

et al., 2003; Murray & Hose, 2005). In addition, several traits that are known to influence taxon survival positively during times of background extinction are found to be irrelevant during mass-extinction episodes, but wide geographical distribution appears to be a universal insurance against extinction (Jablonski, 1986a, 2005; Jablonski & Flessa, 1986; Jablonski & Raup, 1995; Foote, 2003; Kiessling & Baron-Szabo, 2004).

Except for a few studies on Cretaceous marine molluscs (Jablonski, 1986b, 1987), most discussion of general relationships between extinction risk and geographical distribution has focused on modern terrestrial organisms. Geographical ranges and abundance patterns can be determined with much higher precision and accuracy for modern than for fossil taxa. However, temporal patterns of extinction and origination can be assessed directly from the fossil record, whereas extinction risk is usually inferred indirectly in neontological studies, sometimes with circular reasoning (Gaston, 2003). The fossil record also has ample data on originations of new taxa, which are not directly accessible in neontological studies. New data from the marine fossil record can thus substantially add to the assessment of evolutionary implications of geographical distribution.

Here we use a large, taxonomically vetted data set of Triassic–Jurassic benthic marine invertebrates to: (1) explore range-size distributions and their relationship to abundance distributions, (2) assess the relationship between geographical ranges and stratigraphic ranges, (3) analyse the long-term diversity dynamics of narrowly and widely distributed taxa, with special reference to the end-Triassic mass extinction, and (4) test if geographical range size or abundance is more important in determining extinction risk. To overcome some of the problems associated with the incompleteness of the fossil record, we apply modern techniques of sampling standardization.

METHODS

Data and stratigraphic binning

Taxonomic occurrence data of Early Triassic (Induan) to Early Cretaceous (Hauterivian) age were downloaded from the Paleobiology Database (<http://paleodb.org>) on 26 February 2006. The data were taxonomically vetted and filtered to include only valid species and genera of marine macrobenthic and mesobenthic organisms as described by Kiessling *et al.* (2007). Taxa only occurring once in the entire data set were omitted prior to analysis. The final data set consists of 39,251 taxonomic occurrences with 1456 genera and 6019 species in 5880 collections. More than 50% of all occurrences are bivalves, but only about a quarter of genera are bivalves (Table 1). Phylogenetic relationships are poorly known for most taxa under consideration, and we are thus compelled to use a taxic approach. Although most neontological analyses on geographical ranges refer to species, our analyses usually refer to genera because first, with the exception of bivalves, our

Table 1 Distribution of genera, species, taxonomic occurrences and geographical ranges among higher taxa.

Class/phylum	Genera	Species	Occurrences	Median mean geographical range (km)*
Bivalvia	383	2441	22,178	1707
Brachiopoda	267	864	5058	549
Anthozoa	218	1002	4053	1200
Gastropoda	200	697	3629	340
Echinodermata	122	350	1458	344
Porifera	136	345	1234	1051
Others†	130	320	1641	444

*Only for genera with > 0 mean geographical range.

†Bryozoa, Polychaeta, Arthropoda, calcareous algae, problematic organisms, Scaphopoda, Polyplacophora, Hydrozoa.

taxonomic vetting concentrated on the correction of genus names of species occurrences; and second, the stratigraphic ranges of species are usually too small to permit meaningful statistical tests on the fairly coarse stratigraphic resolution we are compelled to use. Only for Jurassic bivalves are the taxonomic data sufficiently clean, and species durations sufficiently long, to allow us to perform analyses at the species level. We use the example of bivalve species to compare basic patterns between taxonomic levels.

Substage ages were utilized for the assessment of taxon longevities. However, the stratigraphic resolution of all analyses of biodiversity dynamics is to the stage level, except for the Early Triassic, for which two stages were combined (Induan and Olenekian) to increase sample size. Our stratigraphic intervals (bins) have an average duration of 6 million years (Myr) in the time interval investigated. There is large variation in the duration of bins (3.1 Myr for the Hettangian to 12.9 Myr for the Norian), but this variation does not pose problems for the statistical tests applied here. We have therefore refrained from lumping particular bins to reduce the variation of interval durations. A finer resolution than stage level is currently not feasible, as too few data would be available for meaningful statistical tests.

Determination of geographical distribution

Geographical distribution consists basically of two variables, which are closely connected: geographical range and abundance. Both variables are apparently so closely linked that one is often regarded as a proxy of the other (Darwin, 1859; Brown, 1995; McKinney, 1997; Gaston, 2003). This, however, may be partially due to the measure of geographical range being dependent on the measure of abundance. If we measure, for example, the geographical range of a species by the number of localities/sites or grid cells in which it is recorded, the number of occurrences serves as a proxy for both abundance and geographical range. In an attempt to have no *a priori* mixture of range size and abundance, we utilize the maximum great circle distance as a measure of geographical range size.

The great circle distance D of two points with given palaeogeographical coordinates is calculated by:

$$\cos(D) = \sin(A) \times \sin(B) + \cos(A) \times \cos(B) \times \cos(L) \quad (1)$$

where A is the palaeolatitude of point a , B is the palaeolatitude of point b and L is the difference between palaeolongitudes of a and b ; all values are in radians. D is thus the arc(cos) of this expression. The palaeocoordinates are calculated automatically upon download from the Palaeobiology Database. The underlying rotation file is based on that of C. Scotese (pers. comm., 2001). The great circle distances between all geographically distinct occurrences of a taxon were measured. The number of distances (n_D) is thus a quadratic function of sample size:

$$n_D = 0.5n^2 - 0.5n \quad (2)$$

where n is the number of geographically distinct occurrences of a taxon. A potential pitfall in this approach is the issue of geographical barriers. The great circle distance always measures the bee-line between two points, irrespective of land areas and deep oceans on this route. Statistically, however, this is probably not relevant on the global scale, because we intend to separate narrow-ranging and wide-ranging taxa rather than to reconstruct migration routes. The precision of palaeogeographical reconstructions does not permit the reliable identification of barriers, and the actual migration routes of fossil taxa remain largely unknown (Aberhan, 2001).

The relative time of maximum dispersal of a taxon (D_t) was calculated as:

$$D_t = (FAD - MRD)/(FAD - LAD) \quad (3)$$

where FAD is the first appearance datum, LAD is the last appearance datum and MRD is the datum of maximum geographical range (all values in Myr). To reduce edge effects, all analyses were limited to genera the last occurrence of which was before the Valanginian. Abundance is usually measured by total population size or population density in ecological studies. A number of issues are involved here (Symonds & Johnson, 2006), and the situation is much worse for the fossil record. The fossil record inevitably preserves only a small fraction of individuals, even for well skeletonized taxa. However, the relative abundances of hard-shelled individuals are often reliably preserved (Kidwell, 2001). Two proxies of abundance are commonly used for fossils. The more direct one is the number of individuals counted at a collection site. The relative counts of durable shells for each taxon are thought to reflect abundance patterns of the living community at this site (Kidwell & Flessa, 1996; Kidwell, 2001), especially for molluscs, and if time averaging was modest (Fürsich & Aberhan, 1990). This type of count is thus a good proxy for local abundance. We determined the mean and maximum percentages of taxa in quantitative collections to determine patterns of local abundance.

As counts are hardly available for all occurrences of a taxon, it is common practice to use as a rough proxy of abundance the number of sites where a taxon has been recorded, that is, the number of its occurrences in individual collections. It has

been shown repeatedly in high-quality data sets, both living and fossil, that the average local abundance of a species is indeed strongly correlated with its number of occurrences (Buzas *et al.*, 1982; Hayek & Buzas, 1997; Buzas & Culver, 1999), so that the global number of occurrences may be used as a surrogate of global abundance. If not otherwise indicated, we refer to this metric when referring to abundance patterns.

Although we also used our measurements of distances to assess patterns of spatial aggregation and range fragmentation, we refrain from reporting these analyses. Range fragmentation is thought to be of paramount importance in modern extinctions (Fagan *et al.*, 2005), but our data set is probably still too small to isolate biologically meaningful patterns. Fossils can be recorded only where sedimentary rocks are accessible and where palaeontologists are performing research. While, in the time interval investigated, countries with a good number of fossil collections are spread globally from New Zealand to Canada and from Chile to Japan, there are some zones in between where fossil data are sparse. This may affect all our measures of geographical distribution, but especially our assessment of spatial aggregation and range fragmentation.

Sampling standardization and biodiversity dynamics

Given the incompleteness of the fossil record, the true stratigraphic and geographical ranges of fossil taxa cannot be determined with 100% confidence. Estimates of these ranges are strongly dependent on preservation and sampling intensity (both summarized under the term 'preservation', below). The stratigraphic ranges of taxa become artificially truncated when preservation worsens (Foote & Raup, 1996). Likewise, taxa with restricted geographical ranges tend to have shorter observed stratigraphic ranges, even if the actual longevities of the taxa are the same as those with broader ranges (Russell & Lindberg, 1988).

There are two options for dealing with this problem. The first is to acknowledge the incompleteness of the fossil record by calculating confidence intervals on observed ranges. While, to our knowledge, confidence intervals have been applied only to stratigraphic ranges (Strauss & Sadler, 1989; Marshall, 1994), there is no reason why confidence intervals could not be applied to geographical ranges as well, even though the spatial component adds another dimension.

The second option is to choose a random subset of the data for the analyses, thus degrading the available fossil record to the same level (that is, to the same number of occurrences or collections). Inevitably, both the geographical ranges within a time interval and the stratigraphic ranges will be lower than in the raw data, but the important bias of unequal sample sizes is minimized. This subsampling approach was utilized here, because it is the aim of this study to compare biologically meaningful patterns rather than to find true absolute geographical and stratigraphic ranges. Several methods for subsampling have been devised (Alroy *et al.*, 2001; Bush *et al.*, 2004). These methods differ in the entities of random draws (occurrences vs. collections) and weighting exponents. We

Table 2 Stratigraphic intervals and number of taxonomic occurrences utilized in this paper.

Interval	Age at base (Myr)	Taxonomic occurrences
Early Triassic	251	727
Anisian	245	2554
Ladinian	237	917
Carnian	228	1804
Norian	216.5	2268
Rhaetian	203.6	2059
Hettangian	199.6	1219
Sinemurian	196.5	2414
Pliensbachian	189.6	4050
Toarcian	183	2900
Aalenian	175.6	772
Bajocian	171.6	1243
Bathonian	167.7	2051
Callovian	164.7	3565
Oxfordian	161.2	4958
Kimmeridgian	155	2214
Tithonian	150.8	3622
Berriasian	145.5	755
Valanginian	140.2	798
Hauterivian	136.4	804

applied a subsampling method known as simple rarefaction, which repeatedly draws a random set of taxonomic occurrences from the global occurrence pool without weighting them. The subsampling size (quota) was determined by the lowest number of taxonomic occurrences, which is 727 in the Early Triassic (Table 2). Although stages could be combined to gain a higher quota, most of the statistical analyses require a time series of as many bins as possible, rather than a time series of fewer bins with more data in them.

Our subsampling procedure drew 720 species occurrences from each bin at random. The geographical distribution of each species or genus was then computed for each bin, based on geographically distinct taxonomic occurrences. This procedure was repeated 50 times and the results were averaged. All relevant measures of the temporal development of geographical range and abundance are based on this data set. We used these data to identify geographical ranges and abundances as: (1) a taxonomic trait, and (2) dynamically. The first approach measures the mean and maximum geographical range and abundance during the life span of a taxon, that is, geographical distribution is viewed as if it represented an ecological property of a taxon. The second approach is a dynamic assessment, which relates extinction risk to the specific geographical distribution in individual bins.

For the characterization of geographical distribution as traits, we identified the mean and maximum geographical range and the mean and maximum number of occurrences during a taxon's life span, as well as the standard deviations and volatilities of both geographical range and abundance. Volatility is the temporal variability of a metric in a time series. Although different metrics are sometimes applied in financial

time series (Kiessling, 2006), we measure volatility as the standard deviation of the first differences. The standard deviation of raw values measures the overall variability of the data, whereas volatility puts the variability in an explicit temporal context. Volatility is greater than the raw standard deviation if the values vary widely between successive bins, but the overall spread of data is comparatively modest. In contrast, a relatively small volatility indicates comparatively modest temporal fluctuations. All metrics were based on actual observations, that is, abundance values were taken only if the taxon occurred in at least one collection in a subsampling trial. Geographical ranges were considered only if the taxon was recorded from at least two localities.

The raw occurrence file was linked with the subsampled measures of geographical distribution for each genus. This file was used to determine the frequency distribution of these measures. The median and percentile values of this distribution were determined to separate widespread or common genera from genera that were narrowly distributed or rare. The median was usually identified on the basis of occurrences of genera, rather than for the number of genera, because we intended to have equally large samples for the analyses rather than equally large pools of genera. This practice could introduce a bias because the number of genera may differ greatly between categories. We thus performed tests, which are reported under Results. The separators were then used to assess diversity dynamics, treating the measures of geographical distribution as a trait of each taxon. This required additional subsampling analyses to achieve sample-standardized diversity dynamics by category. The subsampling quota of these separate analyses was adjusted to the minimum number of occurrences in each bin, and varies between 100 and 200 occurrences per bin and category. This second subsampling procedure was repeated 100 times for each trait to calculate diversity dynamics.

For both the total data set and the categorical data sets, the per taxon rates of extinction (E) and origination (O) were measured with the metrics introduced by Foote (2000a) and elaborated by Foote (2003):

$$E = -\ln[N_{bt}/(N_{bL} + N_{bt})] \quad (4)$$

$$O = -\ln[N_{bt}/(N_{Ft} + N_{bt})] \quad (5)$$

where N_{bt} is the number of taxa crossing both the bottom and top boundaries of an interval, N_{bL} is the number of taxa crossing the bottom interval but having their last appearance within the interval, and N_{Ft} is the number of taxa first appearing in the interval and crossing the top boundary of the interval. Rates are not normalized for stage durations. This may underestimate rates in short stages relative to rates in long stages. However, a recent statistical approach has demonstrated that both extinction and origination tend to be pulsed rather than spread randomly through stratigraphic intervals (Foote, 2005). In any case, the comparison of rates is not affected by normalization or non-normalization of bin durations. The dynamic approach used the subsampled

geographical ranges and abundances within the bins. The distribution of geographical ranges and abundances among taxa in the same interval was then tested against the probabilities of extinction within the bins. For this data set, the geographical range was set to zero if the taxon has only one occurrence in an interval.

Statistical tests

As most of our data have non-normal, strongly skewed distributions, we generally apply nonparametric statistical tests. All correlation tests are based on Spearman's rank correlation coefficients. Proportional values (p) were logit-transformed to $[\ln(p/1-p)]$ prior to statistical tests. Geographically widespread and common taxa were separated from geographically restricted and rare taxa, using medians and percentiles of both the number of taxa and the number of occurrences. The statistical comparison of time series of extinction and origination rates is based on the Wilcoxon signed rank test. Within-bin tests for differences in rates are based on the Mann–Whitney U -test, corrected for ties. Error bars generally report ± 1 SE. In subsampling analyses, the standard error is equal to the standard deviation of the mean achieved in all subsampling analyses.

RESULTS

Within-interval geographical distributions

In this section we present only observations that are directly relevant to the topic of this paper. These observations are based on the raw data, because they refer only to patterns within stages and are thus not affected by temporal variations of fossil preservation, which are due to changes in preserved sedimentary rock outcrop, palaeontological labour and ocean chemistry, among others. We first compare our original proxy for geographical range with an alternative proxy, then look at relationships between abundance and geographical range.

Our metric of maximum great circle distance (MGCD) is theoretically independent of other measures of geographical range, such as grid occupation and the number of distinct locations. However, although MGCD measures geographical range only in one dimension, it is strongly correlated with the metric of grid cell occupation, that is, the number of geographical grids in which a taxon is recorded. For a 30° palaeogeographical grid resolution, we achieve $r = 0.866$, $P < 0.001$ (Fig. 1).

The frequency distributions of range sizes for both all genera and Jurassic bivalve species are strongly right-skewed on normal plots (Fig. 2a,c) and left-skewed on log-transformed histograms (Fig. 2b,d). The same basic pattern is also seen for other separate groups, such as articulate brachiopods, corals and sponges. These frequency distributions are very similar to those of a large suite of modern taxa, both terrestrial and marine (Gaston, 1998; Gaston & He, 2002; Hunt *et al.*, 2005). The only true exception appears to be modern regular sea

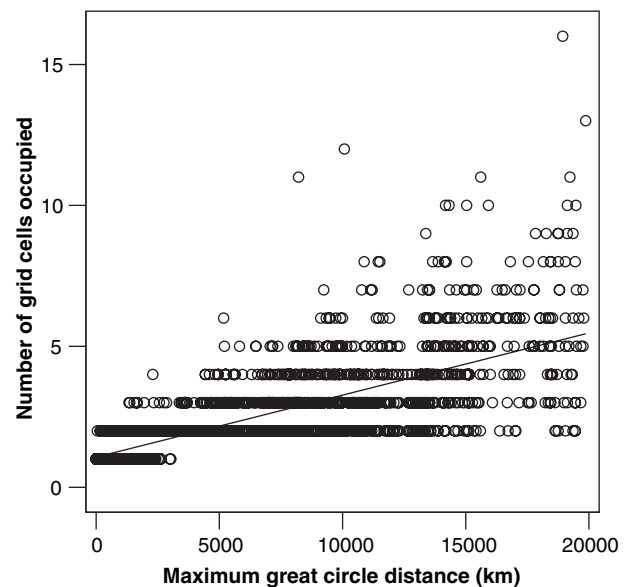


Figure 1 Relationship between two measures of geographical range for Triassic–Jurassic benthic genera. The maximum great circle distances of genera within stages are strongly correlated with the number of 30° palaeogeographical grids occupied by the genera ($r = 0.866$, $P < 0.001$). The fit line is based on a linear regression.

urchins, which have a less skewed distribution (Emlet, 1995). The frequency distributions of abundances are even more right-skewed, even though single occurrences of a taxon in a stage were omitted from the data set (Table 3). In contrast to range-size distributions, the distribution of abundances remains right-skewed on a logarithmic scale. Although there are small differences in the frequency distributions of genera and species, the basic patterns are the same. Thus, although all subsequent analyses in this paper are based on genera, the basic results are probably valid for species as well.

Our analysis of the relationship between abundance and geographical range was first based on local abundance, including all genera that occur in at least three geographically separate quantitative collections in a stage ($n = 722$). Contrary to expectation, there is no significant correlation between MGCD and average or maximum local abundance within collections. This is true for both logit-transformed percentage values (Fig. 3a) and absolute counts of specimens. Individual classes (bivalves, gastropods, articulate brachiopods, anthozoans) show the same basic relationship, with only two exceptions: in bivalves there is a weak positive correlation between the maximum proportional abundance and geographical range ($r = 0.119$, $P = 0.014$, $n = 416$), and in corals (nearly exclusively scleractinians) there is a positive correlation between mean proportional abundance and geographical range ($r = 0.198$, $P = 0.028$, $n = 124$). The correlation between local abundance and geographical range is thus apparently lower than reported in several neontological studies (Hanski *et al.*, 1993; Brown, 1995; Brown *et al.*, 1996).

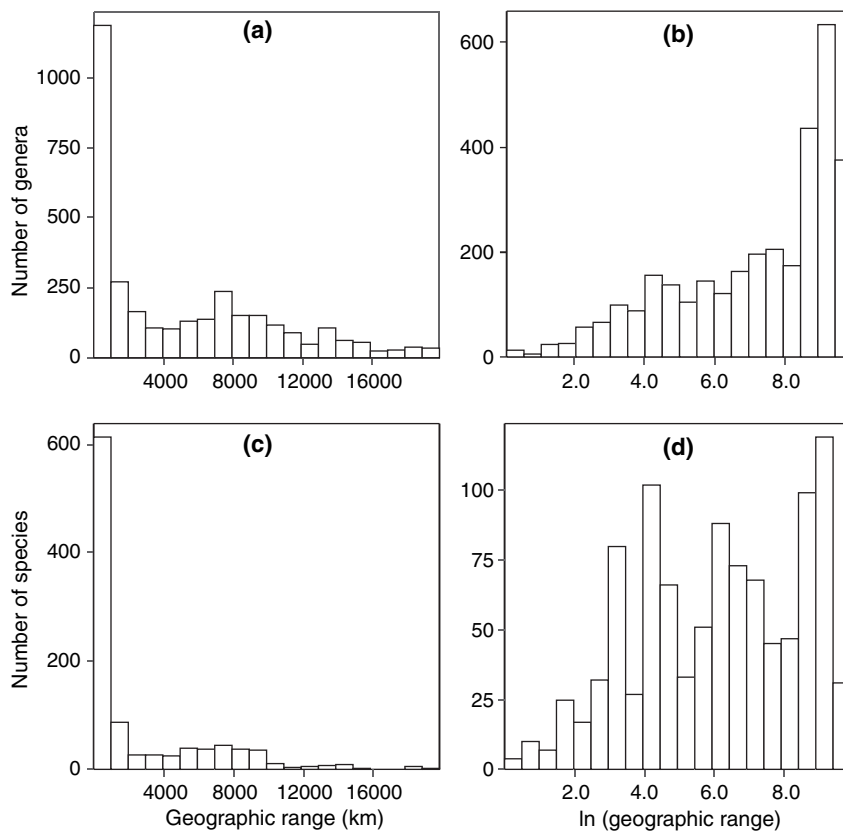


Figure 2 Histograms of geographical ranges by taxon and stage. (a,b) All benthic marine genera; (c,d) Jurassic bivalve species. Note strongly right-skewed distribution of ranges on a normal scale (a,c) and left-skewed distribution on a log scale (b,d). The data set was filtered to include only taxa with at least two geographically separate occurrences.

Table 3 Numerical distribution of range sizes and abundance data.

	Skewness	Kurtosis
Genera*		
Geographical range	0.87	-0.26
Number of occurrences	3.70	21.87
ln (range)	-0.86	-0.33
ln (<i>n</i>)	0.91	0.14
Jurassic bivalve species†		
Geographical range	1.72	2.61
Number of occurrences	3.56	19.35
ln (range)	-0.27	-0.95
ln (<i>n</i>)	1.03	0.49

*3232 data, standard error of skewness = 0.043, standard error of kurtosis = 0.086.

†1024 data, standard error of skewness = 0.076, standard error of kurtosis = 0.153.

However, when taking our indirect measure of global abundance, the total number of occurrences, we find the expected positive correlation between abundance and geographical range ($r = 0.530$, $P < 0.001$, $n = 3232$; Fig. 3b). The discrepancy between the two measures of abundance can probably be resolved when looking at geographical variations in local abundance. Brown (1995) has shown that local abundance varies profoundly within the geographical range of

a species, with highest local abundance being largest near the centre of the geographical distribution. It is also possible that local abundance patterns are more distorted than global abundance patterns. In any case, the general link between abundance and geographical range in our data set is quite similar to some neontological data (e.g. Symonds & Johnson, 2006). The triangular shape of the plots in Fig. 3 suggests that rare and common taxa can have wide distributions, but common taxa do not have restricted geographical ranges.

Body size shows a similar triangular relationship with geographical range in many terrestrial animal species (Blackburn & Gaston, 1996; Brown *et al.*, 1996; Gaston & Blackburn, 1996). We used the geometric mean of height and length of the largest known specimen of Jurassic bivalve species (Aberhan *et al.*, 2006) to test if heterogeneous body-size distributions are likely to introduce a bias to our analyses. There is indeed a significant positive correlation between body size and geographical range ($r = 0.198$, $P < 0.001$, $n = 1005$). However, the correlation is weak, and only 4% of the variance in geographical range is explained by body size. Therefore variations in body size among our taxa are unlikely to affect the results.

Geographical distributions are distributed unequally among higher taxa. Among the higher taxa with more than 100 genera having greater than zero mean geographical ranges, bivalves show the largest median geographical range, followed by corals, sponges, brachiopods and gastropods (Table 1). The rank order of median mean abundances is slightly different.

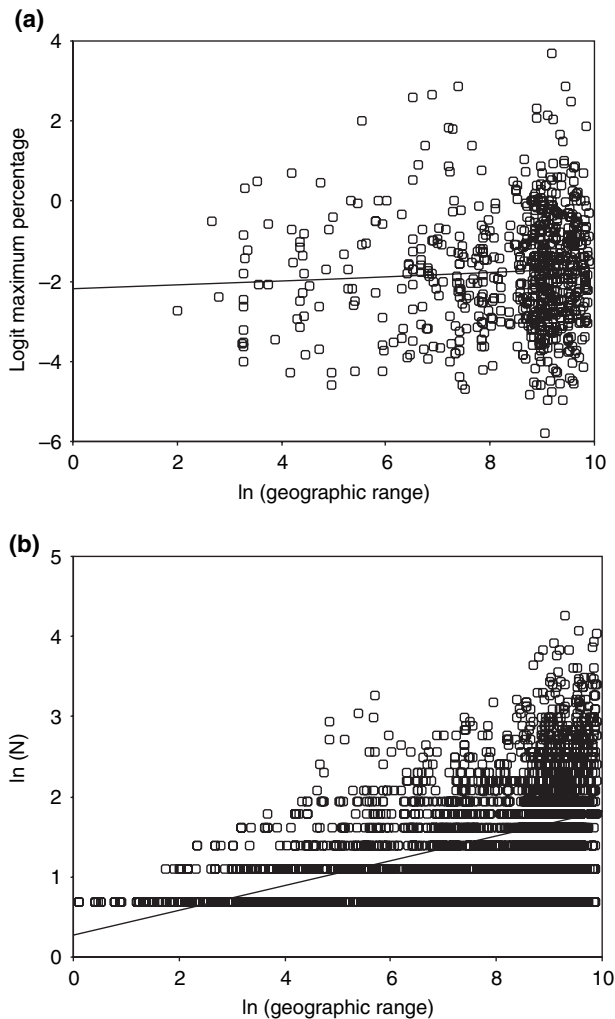


Figure 3 Relationship between abundance and geographical range. (a) Geographical range per stage plotted against the maximum logit-transformed percentages of those genera that appear in at least three taxon-counted samples in that stage. (b) Geographical ranges plotted against the number of geographically separate occurrences, for genera that have at least two geographically separate occurrences in a stage. Fit lines are based on linear regressions. Note that the relationship in (a) is not significant.

Bivalves again have the greatest values, followed by brachiopods, corals, gastropods and sponges.

Temporal distributions of geographical ranges

The simple completeness metric (number of bins in which a taxon is recorded divided by the number of bins between the first and last occurrences of a genus) is 0.76 in the raw data and 0.67 in the subsampled data set, for taxa ranging through at least three bins. Although the completeness is reasonably high, gaps in the record will have major effects on estimates of volatility, especially for long-ranging taxa, because there is a significant inverse correlation between the recorded stratigraphic range and the completeness metric ($r = -0.47$,

$P < 0.001$). This corroborates our approach to limit our measures to those bins in which the genus has actually been recorded.

The volatility of geographical ranges can be assessed when compared with the standard deviation of raw values. In our data set, filtered to include only genera occurring in at least three bins, the standard deviation is significantly higher than the volatility (Wilcoxon signed rank test, $P < 0.001$). This suggests that the volatility of geographical ranges is fairly moderate, and might indicate that there is biological meaning to the fossil geographical range data. Another hint in this direction is the quotient of maximum geographical range to its mean through the life span of a genus. This ratio, determined only for those genera where the maximum is different from the mean, is between 1.1 and 7.4, with a mean of 2.25 ± 0.03 , suggesting that geographical ranges do not usually exhibit a boom-and-bust pattern.

A critical question is also, when in the life span of a taxon is the maximum geographical range (D_t) reached? All analyses here indicate that values of both median and mean D_t are constantly < 0.5 , which indicates that genera tend to have their maximum geographical distribution early in their life span. Histograms of D_t (Fig. 4) suggest that even long-ranging taxa rarely have their maximum geographical distribution late in their life span. The shape of these histograms is similar to the development of the geographical ranges of individual genera (not shown), in that maximum geographical ranges tend to be reached early in the life span of a genus, and ranges tend to shrink gradually towards the extinction of the genus. Although it must be emphasized that this pattern becomes weaker for longer-ranging genera (Fig. 4b), the general pattern is stable: a geologically rapid occupation of close-to-maximum geographical range after origination, a steady-state fluctuation, and finally a gradual decline of geographical ranges towards the end of a lineage. This pattern supports evidence from both modelling (Gaston & He, 2002) and empirical studies (Webb & Gaston, 2000) that maximum range sizes are reached early after speciation. The pattern also suggests that the bias on estimates of first and last occurrences of taxa may be distributed asymmetrically. In non-standardized data, the observed first occurrences might be closer to the true first appearance datum than is the observed last occurrence to the true last appearance datum.

Genus longevity vs. geographical distribution

The central question of whether stratigraphic range is significantly related to geographical range is first addressed by using the total subsampled data set averaged over 50 subsampling trials. The data set was filtered to include only those genera that have a stratigraphic duration of more than 1 Myr and a maximum geographical range > 1 km, and that have their last occurrence prior to the Valanginian stage. The first two filters were applied to reduce the noise introduced by monographic effects, that is, genera that have been described in just one paper in a regional study. The third filter is necessary

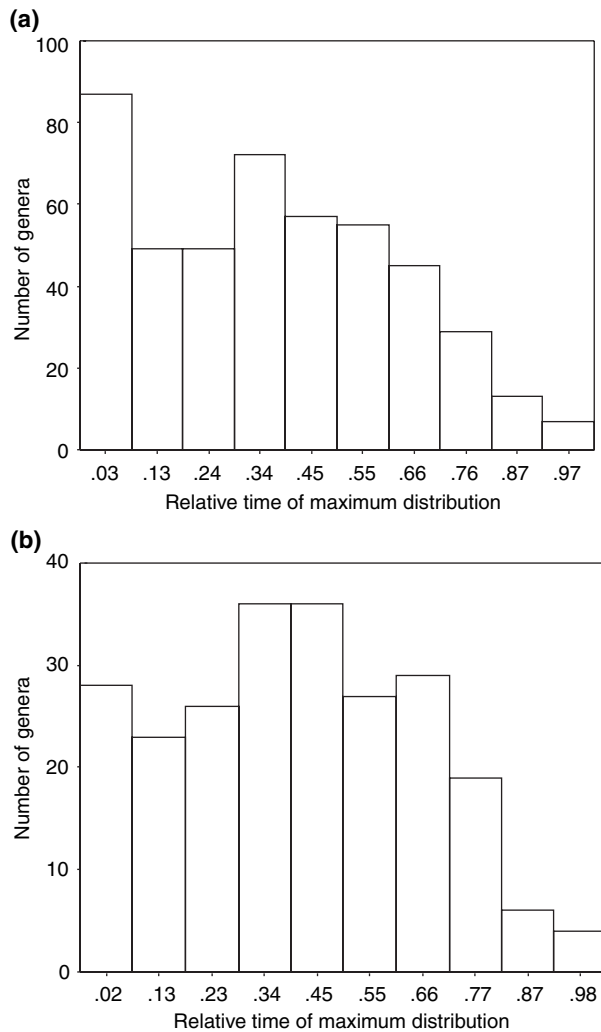


Figure 4 Most genera tend to reach their maximum geographical distribution early in their life span. The histograms show the quantitative distribution of times of maximum dispersal relative to their total stratigraphic persistence (see text for details). (a) Genera that lasted for at least three bins and have their last occurrence before the Valanginian stage. (b) As (a), but genera limited to those lasting for at least five bins.

to reduce the edge effect, that is, to omit taxa that probably range longer into the Cretaceous, but cannot reach longer than to the Hauterivian due to the stratigraphically constrained data set. Due to its devastating effect, the Permian–Triassic mass extinction represents a natural edge at the beginning of the time series, and no filtering was deemed necessary. It should be noted, however, that various additional filters have been applied, but the basic results were as reported below.

Both geographical range and abundance are significantly correlated with taxon longevity (Fig. 5). The strongest rank-order correlation was found between the maximum geographical range that a genus reached in its life span and its stratigraphic range measured in Myr (Fig. 5a, $r = 0.559$, $P < 0.001$, $n = 740$). Maximum values of geographical distribution are thus better predictors of taxon longevity than mean

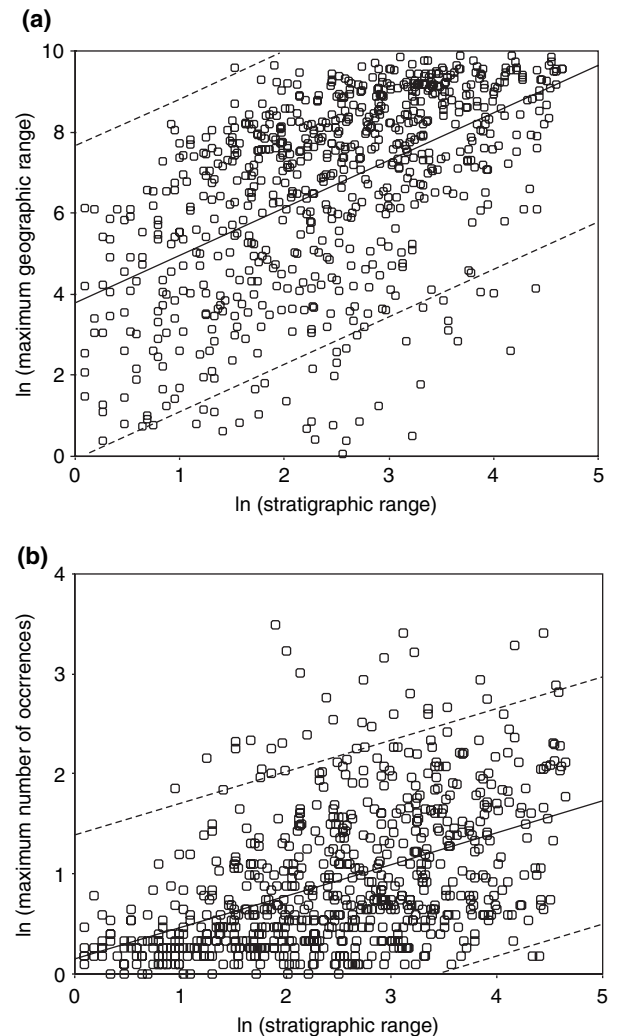


Figure 5 General relationships between the sample-standardized stratigraphic range of genera and (a) maximum geographical range reached in a stratigraphic bin within its life span; (b) maximum number of occurrences reached in a bin.

values, and the correlations are usually higher between geographical range and longevity than between abundance and longevity (Table 4). This suggests that geographical range may be more important than global abundance in determining extinction risk. An interesting feature of the regression of log-transformed values is that the majority of values outside the 95% prediction band are below the lower bound for geographical ranges, but above the upper bound for abundances (Fig. 5). This indicates that much of the residuals are due to genera with unusually large stratigraphic ranges given their geographical range, and unusually small stratigraphic ranges given their global abundance.

Although these results demonstrate the strong influence of both geographical range and global abundance on the longevity of taxa, there are some complicating factors. First, there is a mass-extinction episode in the time series (the end-Triassic mass extinction), which might truncate the stratigraphic ranges of widespread or common taxa. Second, the general

Table 4 Spearman's rank correlations between genus longevities and measures of geographical distribution and abundance.

	Range (Myr)	Range (bins)
Maximum geographical range	0.56	0.42
Mean geographical range	0.44	0.28
Maximum number of occurrences	0.50	0.42
Mean number of occurrences	0.33	0.24

Based on 740 genera in subsampled data set. All correlations significant at $P < 0.001$.

correlations tell us little about potential temporal departures from the general correlation. To assess diversity dynamics through time, we have therefore analysed time series of extinction and origination.

Diversity dynamics related to traits

The overall time series of extinction and origination rates of the subsampled data set are depicted in Fig. 6. The end-Triassic mass extinction is clearly marked by significantly

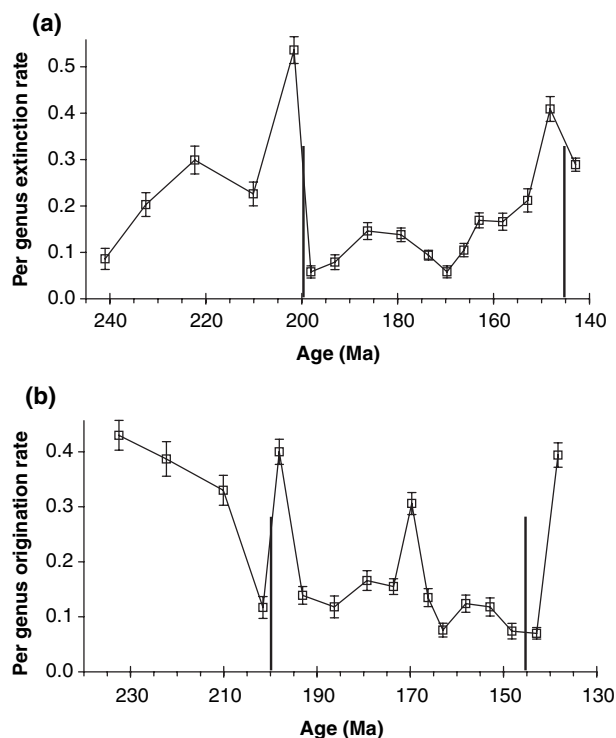


Figure 6 Time series of sample-standardized extinction and origination rates of all benthic marine genera in the Triassic and Jurassic. (a) Extinction rates from the Anisian to Berriasian. (b) Origination rates from the Ladinian to Valanginian. Vertical lines mark the Triassic–Jurassic boundary and the Jurassic–Cretaceous boundary. The subsampling quota was 720 occurrences per stage. Error bars, ± 1 SE, which is the standard deviation of values obtained from all subsampling trials. Extinction and origination rates in this and the following figures were calculated according to equations 4 and 5, respectively.

elevated extinction rates and reduced origination rates (Kiessling *et al.*, 2007). Origination rates declined throughout the Triassic and much of the Jurassic. However, the Jurassic saw two pronounced pulses of origination, one in the Hettangian and the other in the Bajocian. The end-Jurassic extinction is surprisingly high, when compared with analyses of Sepkoski's compendium on stratigraphic ranges (Sepkoski, 1996). As our Triassic and Jurassic rates are otherwise quite similar to the rates of benthic genera in Sepkoski's data set (Sepkoski, 2002), this Tithonian spike requires attention. Edge effects and a change in sampling regimes (from controlled bed-by-bed sampling to outcrop data) may both contribute to this spike. However, even if there is some bias in the estimates of end-Jurassic extinction rates, there is no reason to assume that this bias will differentially affect our categories of geographical distribution.

The separate analyses of evolutionary rates in categories were first determined using as a separator the mean geographical ranges of genera (= their mean maximum great circle distances) in their life span. The median of the mean geographical ranges for occurrences of genera having a greater than zero mean geographical range is 3543 km. This means that half of the total number of genus occurrences contain genera that have a mean geographical range of > 3543 km. These genera were considered widespread, whereas the genera below this threshold were identified as narrowly distributed. Similarly, the median of mean abundances for genus occurrences is 3.1, which allows us to separate common and rare genera. The subsampling quota was 200 occurrences per stage and category. The time series of extinction and origination rates show remarkable differences between widespread and restricted taxa and between common and rare taxa, respectively (Fig. 7). Widely distributed and common genera have significantly lower extinction rates (Wilcoxon signed rank test: $P = 0.005$ and $P = 0.015$, respectively) and lower origination rates ($P < 0.001$ and $P = 0.003$, respectively) than narrowly distributed and rare genera. Just as for the comparisons of longevities, the differences in rates are more pronounced for geographical range than for abundance. In addition, origination rates appear to be more strongly coupled with range and abundance than extinction rates. There are a few exceptions: most prominently in the Rhaetian and Toarcian stages, when widely distributed and common genera have the same or even higher extinction rates than more restricted and rarer genera.

As categories were separated by medians of occurrence counts, and because the distribution of geographical ranges of genera is strongly right-skewed (Fig. 2), many more genera are in the restricted/rare pool than are in the widespread/common pool. The average number of genera is more than twice as high, on average, in the geographically restricted pool. To test if this introduces a bias, we have used the median of geographical ranges among genera, rather than of occurrences. Half the genera have mean geographical ranges of < 872 km and were accordingly defined as geographically restricted. With this separator, genus richness is more equally distributed by definition (the ratio is 0.84, less than one because stratigraphic

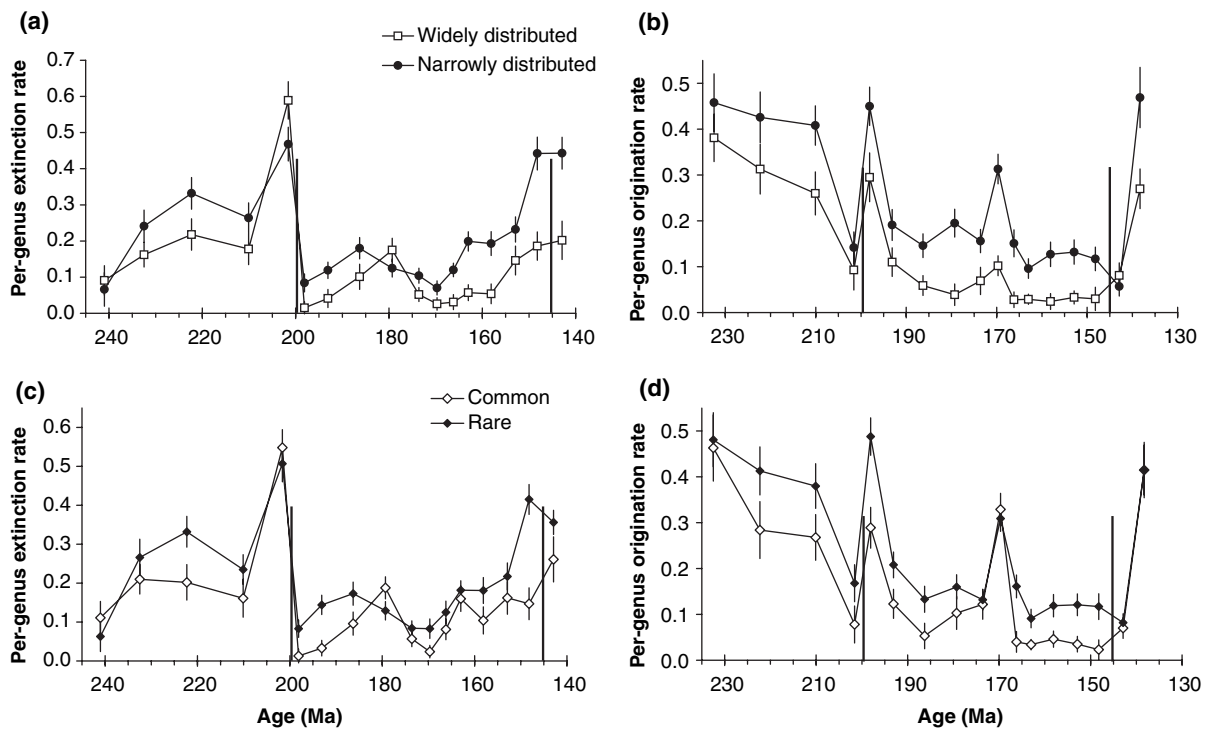


Figure 7 Time series of sample-standardized extinction and origination rates of (a,b) geographically widespread vs. restricted genera and (c,d) common vs. rare genera using the median of means in occurrences as a separator (3543 km and 3.1 occurrences). The subsampling quota was 200 occurrences for each stage and category. Error bars, ± 1 SE.

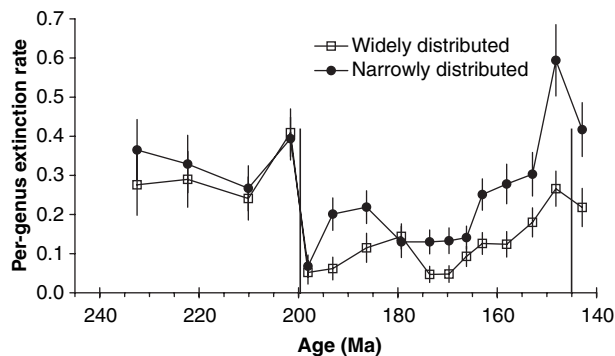


Figure 8 Time series of sample-standardized extinction rates of geographically widespread vs. restricted genera using the median of genus counts as a separator (872 km, quota 100 occurrences). The trajectory starts in the Ladinian, because the Early Triassic 'narrow' category did not reach the subsampling quota. Error bars, ± 1 SE.

singletons are omitted from the analysis), but the number of occurrences is then strongly concentrated in the geographically widespread category. We could thus use only 100 occurrences per stage and category for subsampling, and even then lost the Early Triassic bin. Nevertheless, results are essentially the same (Fig. 8). Insufficient sample sizes in some bins prevented a test for abundances.

The random subsampling approach preferentially draws common taxa within each category, and the median separator

may not be sufficient to separate truly rare from common taxa. To accentuate the separation of widespread/common and restricted/rare genera, we used the lower and upper 33 percentiles of genus occurrences for comparison (Fig. 9). Complete time series could be achieved only with a quota of 100 occurrences per bin and category. Truly widespread genera (> 5090 km) show the expected reduced extinction rate (Wilcoxon signed rank test, $P = 0.001$) compared with genera having a restricted distribution (< 2071 km). The intermediate category has significantly lower extinction rates than the restricted category ($P = 0.002$) and higher rates than the widespread category ($P = 0.017$). Probably due to the small sample size, the difference in extinction rates between abundant genera (upper 33 percentile, > 4 mean occurrences per stage) and rare genera (lower 33 percentile, < 2.5 occurrences) is not significant at a 95% confidence level ($P = 0.084$) and there is no detectable difference between abundant and common (middle 33 percentile) genera ($P = 0.602$). Similarly to the results with the median split between categories, differences tend to be more distinct for origination rates. The most widely distributed taxa are also those with the lowest origination rates compared with both the intermediate percentile ($P = 0.007$) and the lower percentile ($P < 0.001$). For abundance data, the results are similar to extinction rates. There is a significantly lower origination rate in abundant compared with rare taxa ($P = 0.022$) and in common compared with rare taxa ($P = 0.001$). However, there is no significant difference between abundant and common genera,

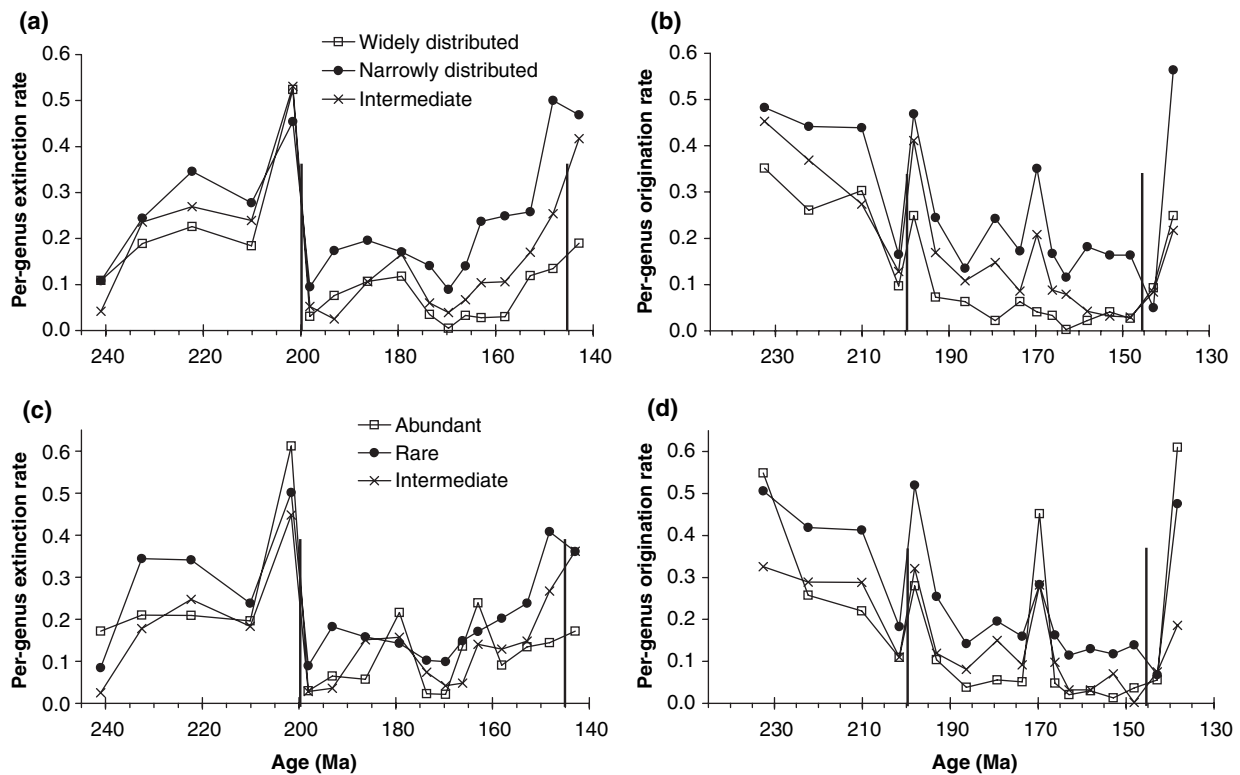


Figure 9 Time series of sample-standardized extinction and origination rates of (a,b) geographically widespread, narrowly distributed and intermediate genera; and (c,d) abundant, rare and intermediate genera using the 33 and 66 percentiles of occurrence counts as separators (2071 and 5090 km, 2.5 and four occurrences; quota 100 occurrences). Error bars are not shown to avoid obscuring details of trajectories.

which is largely due to three outliers associated with generally high origination rates.

To test in more detail the relative roles of geographical range and abundance in determining extinction risk, we have plotted the differences of extinction rates of widespread and restricted taxa against the differences of extinction rates of common vs. rare taxa (Fig. 10). The plots confirm that extinction rates are more closely related to geographical range than to global abundance. They also confirm the exceptional position of the Rhaetian stage, marking the end-Triassic mass extinction.

We have repeated the whole set of analyses with a lower subsampling quota (600 instead of 720) and using the raw data for determining geographical distributions. The basic results are nearly identical. We have also re-run all analyses with zero geographical ranges included. Ranges were otherwise measured only if at least two points per stage were available, while for abundance only one occurrence was needed as a minimum. While volatilities of geographical ranges increase with this treatment, this procedure uses the same protocol for abundances and geographical ranges. The results are also nearly identical, and significance levels are virtually unaffected.

One potential problem needs to be addressed. Bivalves have, on average, significantly larger geographical ranges and abundances than other groups of marine invertebrates in our data set. At the same time, bivalves have long been known to exhibit relatively low evolutionary rates (Simpson, 1944) and our time series of biodiversity dynamics confirm this. Extinc-

tion and origination rates are significantly lower in bivalves when compared with other groups of marine invertebrates (Wilcoxon signed rank test, $P < 0.001$, subsampling quota 180 occurrences). It is thus possible that our finding of elevated extinction and origination rates in narrowly distributed and rare taxa is just a by-product of the lower rates in bivalves, given that bivalves dominate our data set (Table 1). We have analysed just the bivalve data using again the medians of occurrence counts as a separator. The results are nearly identical to those for the complete data set (Fig. 11). Both extinction and origination rates are significantly higher for narrowly distributed than for widely distributed bivalves ($P = 0.002$ and $P = 0.001$, respectively) and for rare than for common bivalves ($P = 0.026$ and $P = 0.004$). One difference is noteworthy: the end-Triassic mass extinction does not represent an outlier in this analysis.

Geographical distribution as a dynamic property

So far, we have addressed only mean and maximum values of geographical distribution. This analysis uses geographical range and abundance as traits of taxa. Additional tests should quantify the relationship between the specific geographical ranges/abundances of taxa in a bin and their probability of extinction in this bin. We performed a series of Mann–Whitney U -tests to see how the observed geographical ranges and numbers of occurrences in a bin are linked with the

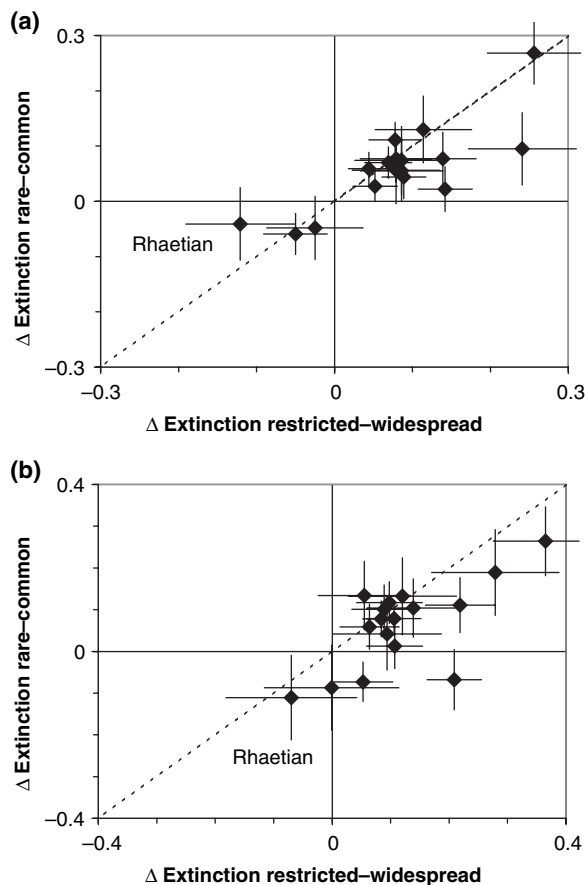


Figure 10 Plots of differences in extinction rates between restricted-widespread and rare-common genera. For values below the dashed line the difference is greater for geographical ranges; for values above the line the difference is larger for abundances. Negative values indicate exceptions from the rule that restricted or rare taxa were more prone to extinction than widespread and common taxa. (a) Comparison for genera with higher and lower than median geographical range and abundance (separators as in Fig. 7). (b) Comparison for genera with 33 and 66 percentile geographical range and abundance (separators as in Fig. 9). Error bars, ± 1 SE.

probability that genera become extinct in this bin. The results suggest that, in most bins, widespread and common taxa show significantly lower extinction risk than restricted and rare taxa (Table 5, columns 2 and 3). Geographical range is more commonly significantly linked with the probability of survival than global abundance (12 vs. nine of 16 bins, significant at 95% CI). The basic results remain stable when singleton genera are included (11 vs. six bins). The absence of significance in some bins is probably due to low extinction rates within the bins (3–13% extinction of non-singleton genera). The end-Triassic mass extinction (Rhaetian stage, 52% extinction) and the Toarcian (16%), however, appear to be true exceptions, when neither a wide geographical range nor global abundance aided survival. In any case, the overall relationship between extinction risk and within-bin geographical range and abun-

dance is strongly significant (Mann–Whitney U -test, $P < 0.001$).

A final test regards the role of temporal changes of geographical distribution in extinction risk. Here we ask, is it more likely for a genus to become extinct if its geographical range and global abundance are lower than normal during its life span? This question was touched upon in Fig. 4, but statistical tests are required for a final judgement. The total sample-standardized data set does indeed show significantly higher probabilities of extinction if the geographical range and abundance are reduced with respect to the mean ($P < 0.001$, values transformed to z scores prior to analysis). When separated by bins, the results are similar to the previous analysis (Table 5, columns 4 and 5), except that now reduced abundances are more often significantly linked to extinction risk than are reduced geographical ranges (13 vs. 11 bins). An additional difference is in the Rhaetian stage, where the temporal development of geographical range and abundance appears to be critical, whereas the within-bin distribution of these measures is not a significant factor.

DISCUSSION

All the results indicate that two important measures of geographical distribution, geographical range and global abundance, are usually strongly linked to extinction risk. Taxa tend to survive for longer, and show reduced extinction rates within time intervals, if they have greater than median geographical ranges and greater than median abundances. The results are independent of the approach, that is, they show the same pattern if geographical distribution is analysed as a trait of the taxon (mean and maximum values of geographical range and abundance over the entire stratigraphic range of a taxon), or if geographical distribution is assessed dynamically (geographical range and abundance within time intervals). Therefore, although it remains disputed if geographical ranges are indeed heritable traits of taxa (Jablonski, 1987; Webb & Gaston, 2003; Hunt *et al.*, 2005), they can be viewed as such for practical purposes. The geographical distribution of a species is determined by a combination of environmental and biological factors (Gaston & He, 2002; Gaston, 2003). Among the latter, niche breadth (Brown, 1984, 1995; Purvis *et al.*, 2000) and dispersal capability (Jablonski, 1986b; Böhning-Gaese *et al.*, 2006) are usually held responsible for determining geographical distribution. The role of dispersal capability has been questioned by a number of studies (reviewed by Gaston, 2003), and we argue that in our data set niche breadth is more crucial than dispersal ability in determining range size. (1) Dispersal capability should especially influence the pace of range occupation, but the geologically rapid occupation of maximum range sizes (Fig. 4) suggests that range occupation is not speed-limited; and (2) although median ranges differ among groups, they all show a similar frequency distribution of range sizes, independent of the presence (most bivalves) or general absence (e.g. articulate brachiopods) of planktotrophic larvae, which favour dispersal ability (Jablonski & Lutz, 1983).

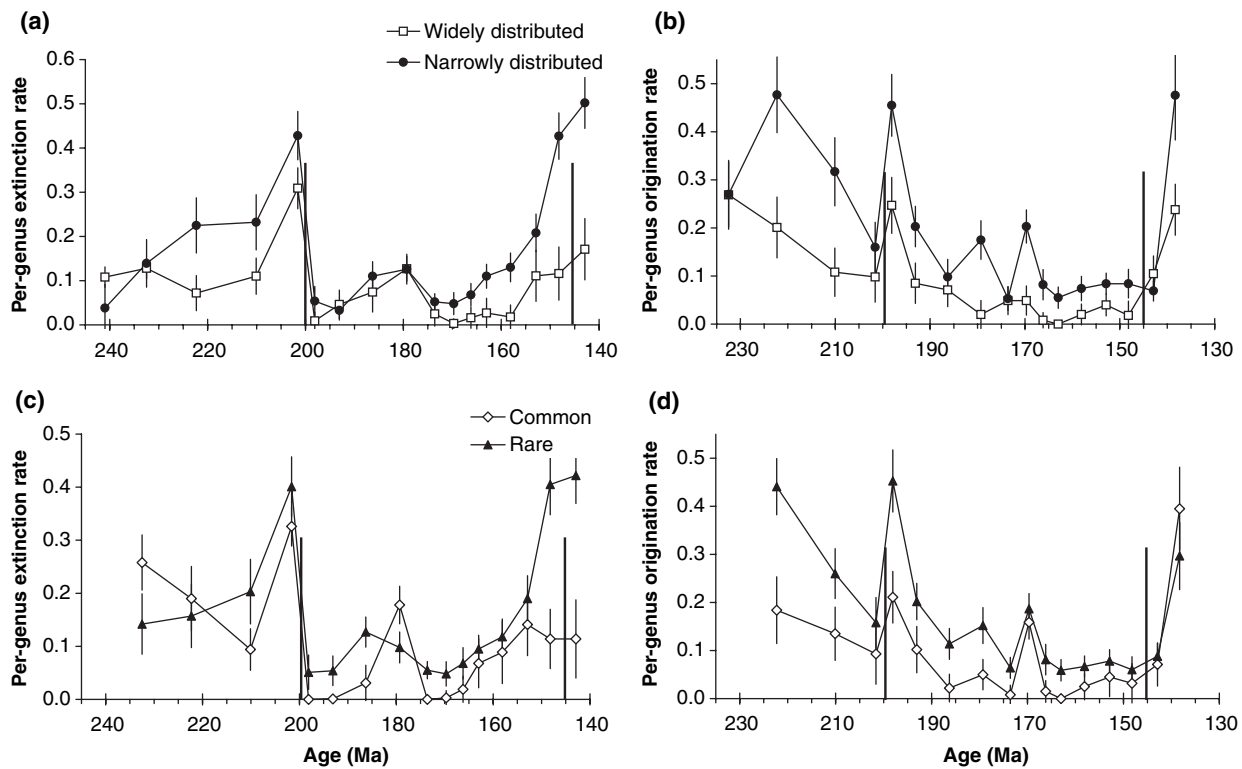


Figure 11 Time series of sample-standardized extinction and origination rates of (a,b) geographically widespread vs. restricted bivalve genera; (c,d) common vs. rare bivalve genera using the median of means in occurrences as a separator (5272 km and 3.7 occurrences). The subsampling quota was 100 occurrences for each bin and category. The trajectories of common vs. rare bivalves start later because the quota was not reached for the Early Triassic 'rare' category. Error bars, ± 1 SE.

Table 5 Results of Mann–Whitney *U*-tests on the dependency of extinction risk of genera from geographical distribution per stage.

Interval	Probability that extinction risk is independent of geographical distribution in interval		Probability that extinction risk is independent of geographical distribution relative to taxon average	
	Geographical range	Number of occurrences	Geographical range	Number of occurrences
Anisian	0.039	0.039	0.062	0.027
Ladinian	0.324	0.394	0.160	0.012
Carnian	0.017	0.018	0.009	0.003
Norian	< 0.001	< 0.001	< 0.001	< 0.001
Rhaetian	0.398	0.697	0.004	0.009
Hettangian	0.018	0.018	0.039	0.012
Sinemurian	0.050	0.101	0.128	0.020
Pliensbachian	< 0.001	0.001	0.022	< 0.001
Toarcian	0.098	0.070	0.011	0.041
Aalenian	0.012	0.097	0.003	0.039
Bajocian	0.066	0.068	0.355	0.776
Bathonian	0.344	0.626	0.178	0.473
Callovian	0.002	0.058	0.024	0.022
Oxfordian	< 0.001	< 0.001	0.001	< 0.001
Kimmeridgian	< 0.001	< 0.001	< 0.001	< 0.001
Tithonian	< 0.001	< 0.001	0.178	0.740
Berriasian	< 0.001	0.004	0.010	0.199

Based on subsampled data set, singletons excluded.

Our assessment of global abundance was based largely on the number of geographically distinct occurrences, and does not provide a measure of local density. Global abundance and geographical range are closely correlated and cannot be separated clearly by our subsampling approach because there are simply too few rare-but-widespread and common-but-restricted taxa to obtain a reasonable sample size. In any case, our results suggest that geographical range *per se* is more important than abundance in determining extinction risk (Fig. 10). A straightforward explanation could be that regional factors dominate over global factors in determining extinction risk. Consider two species with the same number of geographically distinct occurrences, where the first has a global distribution and the second occurs only on the shelves of one palaeocontinent. If regional environmental change affects this one continent but not others, the first species will survive whereas the second will become extinct. If global factors were more important, we would not expect to see a difference, and it would also be difficult to explain why range size should matter at all.

Most of our analyses refer to extinction rates. However, relationships between origination and geographical distribution are equally interesting. Origination rates are consistently lower in widespread and common taxa than in geographically restricted and rare taxa. The same observation was also reported for Late Cretaceous and Recent gastropods (Jablonski, 1986b; Jablonski & Roy, 2003) and Neogene bivalves (Stanley, 1986). Jablonski (1986b) offered a potential mechanism for higher origination rates in lineages with reduced geographical ranges: gene flow between local populations is more likely to be interrupted by habitat alterations than in widespread lineages. This explanation is appealing and could also explain why the differences are generally clearer for originations than for extinctions. If global change is added to regional habitat alteration, the role of geographical distribution in determining extinction will probably be reduced (see the discussion on mass extinctions below), but the mechanisms of speciations (or originations of higher taxa) will not be affected, even if origination rates peak after mass extinctions. In addition, there is a general tendency of Mesozoic and Cenozoic diversity dynamics to be controlled by originations rather than extinctions (Foote, 2000b). The reasons for this origination control are only beginning to be explored, but are perhaps linked to the ecological structure of communities (Wagner *et al.*, 2006).

Many other factors, extrinsic and intrinsic, are correlated with extinction risk (McKinney, 1997; Purvis *et al.*, 2000) and, given that just 31% of the variance in genus longevities is explained by geographical range (based on r^2 values), these factors should contribute substantially to the long-term trajectories. Although it is beyond the scope of this study to discuss even a fraction of these factors, we emphasize the potential bias of taxon ages. A study on extant birds suggested a positive correlation between taxon age and extinction risk of species in this taxon (Gaston & Blackburn, 1997). It may indeed be true that phylogenetic branching increases the

extinction risk of ancestor lineages, but this is not due to taxon age *per se* (Pearson, 1998). In our data set, we find the opposite: old lineages show significantly lower extinction rates than young lineages in a bin. This observation is stable, with different criteria for separating old and young lineages (mean, median, percentiles). This pattern is clearly related to geographical distribution. The old lineages are those that have, on average, the largest geographical range and the greatest abundance in a bin. A similar observation was also made by Böhning-Gaese *et al.* (2006) for extant birds. We suspect that the traits favouring wide distribution also favour survival over longer periods, rather than geographical distribution being an epiphenomenon of taxon age. Our main argument is the geologically rapid occupation of maximum range sizes, which is especially evident in taxa with relatively short stratigraphic ranges (Fig. 4a).

The lack of phylogeny in our approach prevents us from separating true extinctions (the end of a lineage without descendants) from pseudo-extinctions, where phyletic transformation (anagenesis) or bifurcate branching (cladogenesis) results in new taxonomic names without a true biological extinction. We tacitly assumed true extinctions for our extinction rates and budding from ancestral lineages as the dominant process of speciation (or origination of genera). A phylogenetically resolved data set would certainly allow for a deeper insight, but most of the taxa in our data set have not yet been subjected to phylogenetic analyses.

Substantial new insights into the dynamics of mass extinctions can be inferred from our results. The end-Triassic mass extinction shows several interesting departures from the normal pattern. Most importantly, the overall tendency for taxa with restricted ranges and low abundance to be more prone to extinction disappears, or is even reversed, in the Rhaetian stage. This result is surprising considering that large geographical range is often seen as one of the few traits that universally supports survival both during normal background extinction intervals and during mass-extinction episodes (Jablonski, 1986a, 2005; Westrop, 1989; Jablonski & Raup, 1995; Kiessling & Baron-Szabo, 2004; but cf. Smith & Jeffery, 1998 for an alternative view), whereas abundance is not (Lockwood, 2003; Kiessling & Baron-Szabo, 2004). Previous rigorous studies have focused largely on the Cretaceous background extinctions and the end-Cretaceous mass extinctions, or were limited to just mass-extinction episodes, for which Jablonski (2005) listed several references. Few studies compared the relationship between extinction rates and geographical distribution over prolonged times in other periods, and these yielded equivocal results. In the Permian, Erwin (1996) noted significantly higher extinction rates of narrowly distributed gastropods in the Wordian and Capitanian stages, but no significant difference in the end-Permian (Changhsingian) mass extinction. Foote's (2003) analysis of the end-Ordovician mass extinction also suggest a reduced role of geographical distribution in aiding survivorship, because very widespread genera (those found on seven palaeocontinents) had similar extinction rates to those for genera found

on only two or three palaeocontinents, even though extinction rates are generally inversely correlated with geographical distribution in the Ordovician. We have shown above that individual higher taxa may behave differently. Bivalves exhibit the regular pattern of narrowly distributed genera being affected more strongly by extinction than widely distributed genera, as reported previously by Hallam (1981). Given that few other studies have analysed a large suite of taxa in conjunction, we are unable to judge if the end-Triassic extinction is unique in its biological selectivity regime, or if other mass-extinction episodes will turn out to have been similar in this regard. Interestingly, the much smaller extinction spike in the Toarcian shows the same reversal, with widely distributed and common taxa being at least as much affected as narrowly distributed and rare genera. The end-Jurassic extinction spike, however, does not depart from the general pattern. We have already indicated our concerns regarding the magnitude of the end-Jurassic extinction as calculated from our data set, and the pattern suggests that no qualitative difference separates the Jurassic–Cretaceous boundary from background extinctions. The causes of the end-Triassic mass extinction and the Toarcian extinction are still discussed, but both events are probably related to global factors, climatic and/or oceanographic (Aberhan & Baumiller, 2003; Kiessling *et al.*, 2007). The pattern might thus indicate that the geographical selectivity of extinction disappears when global factors became dominant over regional factors in habitat alteration.

CONCLUSIONS

The fossil record is able to provide answers and add new perspectives to questions regarding the evolutionary implications of geographical distribution, if heterogeneous sampling is balanced by standardization. We have shown that the frequency distributions of range sizes of fossil marine benthic invertebrates are similar to those of extant organisms. Although local abundance is poorly correlated with geographical range, global abundance measured by the number of geographically distinct occurrences is a good predictor of geographical range measured by the maximum great circle distance of genus occurrences in an interval of time. The maximum geographical range of a lineage tends to be reached rapidly after origination and, after a period of steady-state stability, ranges tend to be eroded towards the end of a lineage's life span.

Both geographical range and global abundance are important predictors of taxonomic longevity. This is true if geographical distribution is evaluated as a trait, or if it is assessed dynamically. The dynamic assessment shows that in most intervals of time, taxa have a significantly greater risk of extinction if they have restricted geographical ranges and are rare relative to other taxa in that bin. In addition, in most time intervals taxa tend to become preferentially extinct if the geographical ranges and abundances are reduced relative to the mean values during their life span. This is not trivial, because there are substantial fluctuations of range sizes during the life

span of almost every taxon. When viewed as a trait, significantly reduced extinction rates and origination rates are evident in taxa that tend to be widespread and common, compared with taxa that tend to be narrowly distributed and rare. Although geographical range and global abundance are closely related, our analyses suggest that geographical range in isolation is more important than abundance in isolation. This pattern can best be explained by the greater impact of regional habitat alterations on the survival of geographically restricted taxa, and the greater probability of gene flow interruption and hence allopatric speciation in restricted taxa.

During the end-Triassic mass extinction, neither geographical range nor abundance was relevant in determining extinction risk. Even though this pattern is prominent only when all taxa are analysed in combination, this offers a new perspective on extinction risk, which has relevance to macroevolutionary studies and perhaps to conservation biology. The pattern might indicate that, if global factors become dominant over regional factors in causing extinctions, it becomes irrelevant whether a taxon is widespread or geographically restricted, common or rare.

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